7030 DPS

File No. JA BP6A

MEMORY PATTERN TESTS

JA BP 6A

May 5, 1961

1. Programs becoming obsolete - JA CP6 Memory Pattern Tests.

2. Used to provide pattern generating tool for memory scoping.

1. PURPOSE

The MPT 06 I and II programs were written to provide a means of generating patterns within memory, facilitating scoping of memory circuits. Part I checks four 16K B memory boxes starting in location 100000_8. Part II checks two 16K A memory boxes, starting in location 40_8.

2. PROGRAM INTRODUCTION

This program operates continuously with no error indications. It relies on scoping to detect errors.

3. OPERATING PROCEDURES

1. Loading Procedures

   1.1 PUNFUL Binary deck consists of 7 cards.

   1.2 Procedure

      1.2.1 Push Master Reset

      1.2.2 Place Interrupt disable in the active position.

      1.2.3 Place maintenance switch in active (Maintenance Mode) position.

      1.2.4 Place sense switches in desired positions.

      1.2.5 Depress Initial Program Load on maintenance console.

      1.2.6 Place deck in card reader and push START.

2. Options

Sense Switch (Maintenance Bits)	Effect of setting to a 1
35	Cycle on one pattern word.
36	Cycle on test of upper memories
37	Cycle through patterns repetitively in lower memories.
38	LOOP-T-LOOP

4. PROGRAM PHILOSOPHY

Six pattern words are used.

1) All zeros word
2) All ones word
3) Alternating one and zero (101010 ...)
4) Alternating zero and one (010101 ...)
5) Zeros ECC word
6) Ones ECC word

Each word is stored throughout memory, either upper or lower. The operator may cycle on any pattern word or sequence through them. He may cycle on either the A or the B memories, or may alternate between them.

PROGRAM WRITEUP ADDENDUM

Program **Mem Pattern**

File No. **JA BP6A**

MAINTENANCE TAPE CONTROL CARD

Location/s of Exit Branch/es

1.	**1**	**0**	**0**	**0**	**1**	**1**	.	**0**
2.							.	
3.							.	
4.							.	
5.							.	
6.							.	

Pre-Loading Manual Intervention Required ? Yes* _____ No **x**

Pre-Loading Procedure (If Any)

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PRNID,MEMORY PATTERN %6 BOX□ A. BROWN
PRNID,MAY 5, 1961, JA BP6A
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```
                SEM,6,G
                PUNFUL,
                SLC,509.                                                                                    000775.00
                CW%CDSC¤,510.,2,510                                                  776.00 60 000040.01 FE 000775.00
                CW%CDSC¤,512.,FINM1-MPATA,$&1.0                                     1000.00 60 001520.01 FF 000776.00
                CW%CD¤,32768.,FINM2-MPATB,$                                         100000.00 20 022320.01 FF 000777.00
                SLC,512.                                                                                    001000.00
MPATA           LCI,$X2,50                                                           62.05 02               001000.00
                      NOP,0.0       @SET UP TO PROTECT LOW MEM TEST                 0.30 00                001000.40
                T,$X2,MPATB,MPST1   @PUT LOW MEM TEST IN STORAGE                    100000.00 80 001034.04 20 001001.00
MPATA1          LX,$X1,MPAXW1                                                        1013.02 10             001002.00
                      LCI,$X2,6     @SET UP TMT CNT & PATTERN CNT                   6.05 02                001002.40
MPAT1           TI,1,MPTDA1%$X1¤,32768.  @PLACE PATTERN IN TEST AREA                1014.00 81 100000.02 A0 001003.00
                T,$X1,32768.,32769.  @TMT PATTERN THROUGH MEM                       100000.00 80 100001.02 20 001004.00
                BB,4.35,MPAT1       @TURN SW 35 ON TO LOOP PATTERN                  4.43 80 001003.34 02   001005.00
                V&I,$X1,1.0                                                          1.03 05                001006.00
                      CB,$X2,MPAT1  @CHANGE PTRN, & BR TO USE                       1003.04 48             001006.40
                BB,4.36,MPATA1      @TURN SW 36 ON TO LOOP                          4.44 80 001002.34 02   001007.00
                LCI,$X2,50.                                                          62.05 02               001010.00
                      NOP,0.        @SET UP TO TMT LOW MEM TEST                     0.30 00                001010.40
                T,$X2,MPST1,MPATB   @RESTORE LOW MEM TEST                           1034.00 80 100000.04 20 001011.00
                B,MPATB                                                             100000.10 00           001012.00
                      NOP,0         @BRANCH TO LOW MEM TEST                         0.30 00                001012.40
MPAXW1          %W,0,65535,$                                                         0.00 03 777760.02 0B   001013.00
MPTDA1          %8¤DD%BU,64,8¤,0      @0S                                        000000000000000000000  001014.00
MPTDB1          %8¤DD%BU,64,8¤,1 777 777 777 777 777 777 777 777 @1S             17777777777777777777   001015.00
MPTDC1          %8¤DD%BU,64,8¤,1 252 525 252 525 252 525 252 @1010 PATTERN       12525252525252525252   001016.00
MPTDD1          %8¤DD%BU,64,8¤,0 525 252 525 252 525 252 525 @0101 PATTERN       05252525252525252525   001017.00
MPTDE1          %8¤DD%BU,64,8¤,0 000 000 000 060 000 000 001 @0S ECC             00000000000600000001   001020.00
MPTDF1          %8¤DD%BU,64,8¤,1 777 777 777 757 777 777 777 @1S ECC             17777777775777777777   001021.00
                %8¤DD%BU,64,8¤,0,0,0,0,0,0,0,0,0,0,0 @BUFFER                     000000000000000000000  001022.00
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```
MPTDE2  %8□DD%BU,64,8□,0 000 000 000 060 000 000 001 (OS ECC          0000000000060000000001    100017.00
MPTDF2  %8□DD%BU,64,8□,1 777 777 777 757 777 777 777 @IS ECC          1777777777757777777777    100020.00
        %8□DD%BU,64,8□,0,0,0,0,0,0,0,0,0,0      @BUFFER                00000000000000000000    100021.00
                                                                       00000000000000000000    100022.00
                                                                       00000000000000000000    100023.00
                                                                       00000000000000000000    100024.00
                                                                       00000000000000000000    100025.00
                                                                       00000000000000000000    100026.00
                                                                       00000000000000000000    100027.00
                                                                       00000000000000000000    100030.00
                                                                       00000000000000000000    100031.00
                                                                       00000000000000000000    100032.00
MPST2   DR%BU,64,8□,562                                  1062.00                               100033.00
        CNOP,0,0
FINM2   NOP,0,0                                            0.30 00                            101115.00
            NOP,0,0                                         0.30 00                            101115.40
        END,MPATA                                        1000.00                               101116.00
```